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DEPARTMENT OF CSE(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

DATABASE MANAGEMENT SYSTEMS LABORATORY

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EXERCISE –1: CREATION OF TABLES **Date:30/07/2026**

Aim:
To create database tables using SQL CREATE TABLE statements with appropriate columns, data types, and constraints for storing and managing data efficiently in a relational database management system (RDBMS).

1) Create a table called employee for the following structure.

Name	Type
Empno	Number
Ename	Varchar2(20)
job	Varchar2(20)
Mgr	Number
Sal	Number

1. Create Employee Table :

```
CREATE TABLE Employee (
Empno NUMBER PRIMARY KEY,
Ename VARCHAR2(20),
Job VARCHAR2(20),
Mgr NUMBER,
Sal NUMBER
);
```

1(a). Add a Column Commission

```
ALTER TABLE Employee
ADD Commission
NUMBER;
```

1(b). Insert Any Five Records

```
INSERT INTO Employee VALUES (10, 'RAVI', 'MANAGER', NULL,
60000, 5000); INSERT INTO Employee VALUES (11, 'PRIYA', 'ANALYST',
10, 45000, 3000);
INSERT INTO Employee VALUES (12, 'KIRAN', 'CLERK', 10, 25000, 1000);
INSERT INTO Employee VALUES (19, 'SURESH', 'SALESMAN', 11, 30000, 2000);
```

INSERT INTO Employee VALUES (20, 'ANITHA', 'ACCOUNTANT',10, 40000, 2500);

Verify the records:

SELECT * FROM Employee;

1(c). Update the Job Column Details

Example: Change the job of employee 12 from CLERK to SENIOR CLERK.

UPDATE Employee

SET Job = 'SENIOR

CLERK' WHERE

Empno = 12;

Or change another employee:

UPDATE Employee

SET Job = 'HR

MANAGER' WHERE

Empno = 20;

Display updated records:

SELECT * FROM

Employee;

1(d). Rename the Column Using ALTER Command

Suppose we rename **Sal** to **Salary**.

ALTER TABLE Employee

RENAME COLUMN Sal TO Salary;

Describe the table:

DESC Employee;

1(e). Delete the Employee Whose Empno is 19

DELETE FROM

Employee WHERE

Empno = 19;

Verify:

SELECT * FROM Employee;

---

2) Create a department table with the following structure.

Name	Type
Deptno	Number
Deptname	Varchar2(20)
Location	Varchar2(20)

a. Add column designation to the department table.
b. Insert values into the table.
c. List the records of emp table grouped by deptno.
d. Update the record where deptno is 9.

Create Department Table

```
CREATE TABLE Department (
   Deptno NUMBER,
   Deptname VARCHAR2(20),
   Location VARCHAR2(20)
);
```

a. Add column Designation to the Department table

```
ALTER TABLE Department
ADD Designation VARCHAR2(20);
```

b. Insert values into the table

```
INSERT INTO Department (Deptno, Deptname, Location, Designation)
VALUES (1, 'HR', 'Delhi', 'Manager');

INSERT INTO Department (Deptno, Deptname, Location, Designation)
VALUES (2, 'Sales', 'Mumbai', 'Executive');

INSERT INTO Department (Deptno, Deptname, Location, Designation)
VALUES (9, 'Finance', 'Chennai', 'Accountant');

COMMIT;
```

c. List the records of EMP table grouped by Deptno

If you want the number of employees in each department:

```
SELECT Deptno, COUNT(*) AS Total_Employees
FROM Emp
GROUP BY Deptno;

SELECT * FROM Emp
ORDER BY Deptno;
```

d. Update the record where Deptno = 9

Example:

```
UPDATE Department
SET Location = 'Bangalore'
WHERE Deptno = 9;
COMMIT;
```

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3) Create a table called branch table.

Branch Table Structure

Name	Type
Branch_name	VARCHAR2(20)
Branch_city	VARCHAR2(20)
Assets	NUMBER

a. Increase the size of data type for asserts to the branch.
b. Add and drop a column to the branch table.
c. Insert values to the table.
d. Update the branch name column.
e. Delete any two columns from the table.

SQL Structure:

```
CREATE TABLE Branch (
   Branch_name VARCHAR2(20),
   Branch_city VARCHAR2(20),
   Assets NUMBER
);
```

a. Increase the size of data type for Assets

Increase the precision of the Assets column.

```
ALTER TABLE Branch
MODIFY Assets NUMBER(12);
```

Or specify precision and scale if required:

```
ALTER TABLE Branch
MODIFY Assets NUMBER(12,2);
```

---

b. Add and Drop a Column

Add a column

```
ALTER TABLE Branch
ADD Branch_Manager VARCHAR2(30);
```

Drop the column

```
ALTER TABLE Branch
DROP COLUMN Branch_Manager;
```

---

c. Insert Values into the Table

```
INSERT INTO Branch
VALUES ('Hyderabad', 'Hyderabad', 5000000);

INSERT INTO Branch
VALUES ('Vijayawada', 'Vijayawada', 3500000);
```

```
INSERT INTO Branch
VALUES ('Visakhapatnam', 'Visakhapatnam', 4500000);

COMMIT;
```

---

d. Update the Branch Name Column

```
UPDATE Branch
SET Branch_name = 'Secunderabad'
WHERE Branch_name = 'Hyderabad';

COMMIT;
```

---

e. Delete Any Two Rows from the Table

Note: SQL cannot delete two **columns** using the DELETE statement. DELETE removes **rows**. If your question says "Delete any two columns," it is likely a typo and should be **Delete any two rows**.

Delete two rows:

```
DELETE FROM Branch
WHERE Branch_name = 'Vijayawada';

DELETE FROM Branch
WHERE Branch_name = 'Visakhapatnam';

COMMIT;
```

If the intention is to drop two columns, use:

```
ALTER TABLE Branch
DROP COLUMN Branch_city;

ALTER TABLE Branch
DROP COLUMN Assets;
```

View the Table

```
SELECT * FROM Branch;
```

---

4. Create a table called sailor table

Name	Type
Sid	Number
Sname	Varchar2(20)
Rating	Varchar2(20)

Create the SAILOR Table

```
CREATE TABLE Sailor
(
   Sid NUMBER,
   Sname VARCHAR2(20),
   Rating VARCHAR2(20)
);
```

---

a. Add Column Age to the Sailor Table

```
ALTER TABLE Sailor
ADD Age NUMBER(3);
```

---

b. Insert Values into the Sailor Table

```
INSERT INTO Sailor VALUES (101, 'Ravi', '7', 22);
INSERT INTO Sailor VALUES (102, 'Kiran', '9', 25);
INSERT INTO Sailor VALUES (103, 'Suresh', '8', 24);
INSERT INTO Sailor VALUES (104, 'Mahesh', '10', 27);
```

COMMIT;

---

c. Delete the Row with Rating > 8
Since Rating is of type VARCHAR2, convert it to a number before comparison.
DELETE FROM Sailor
WHERE TO_NUMBER(Rating) > 8;

COMMIT;
Note: If Rating were defined as NUMBER, the query would simply be:
DELETE FROM Sailor
WHERE Rating > 8;

---

d. Update the Column Details of Sailor
Update Sailor Name
UPDATE Sailor
SET Sname = 'Anil'
WHERE Sid = 101;

COMMIT;
Update Rating
UPDATE Sailor
SET Rating = '8'
WHERE Sid = 103;

COMMIT;
Update Age
UPDATE Sailor
SET Age = 26
WHERE Sid = 104;

COMMIT;

---

e. Insert NULL Values into the Table
INSERT INTO Sailor (Sid, Sname, Rating, Age)
VALUES (105, NULL, NULL, NULL);

COMMIT;

Display the Table
SELECT * FROM Sailor;
Expected Output

SID	SNAME	RATING	AGE
101	Anil	7	22
103	Suresh	8	24
105	NULL	NULL	NULL

5. Create a table called reserves table:

Name	Type
Boat id	Integer
sid	Integer
day	Integer

5. Create Reserves Table

```
CREATE TABLE Reserves (
   Boat_id NUMBER(5),
   Sid NUMBER(5),
   Day NUMBER(2)
);
```

Insert 5 Records into the Reserves Table

```
INSERT INTO Reserves VALUES (101, 1, 10);
INSERT INTO Reserves VALUES (102, 2, 12);
INSERT INTO Reserves VALUES (103, 3, 15);
INSERT INTO Reserves VALUES (104, 4, 18);
INSERT INTO Reserves VALUES (105, 5, 20);

COMMIT;
```

---

a. Insert Values into the Reserves Table

```
INSERT INTO Reserves VALUES (106, 6, 25);

COMMIT;
```

---

b. Add Column Time to the Reserves Table

```
ALTER TABLE Reserves
ADD Time VARCHAR2(10);
```

---

c. Alter the Column Day Data Type to DATE

If the table already contains numeric values in the **Day** column, Oracle does not allow changing the datatype directly from NUMBER to DATE. Create the table with DATE initially, or use the following approach:

```
ALTER TABLE Reserves
ADD Reserve_Date DATE;

UPDATE Reserves
SET Reserve_Date = TO_DATE(Day || '-01-2026', 'DD-MM-YYYY');

ALTER TABLE Reserves
DROP COLUMN Day;

ALTER TABLE Reserves
RENAME COLUMN Reserve_Date TO Day;

COMMIT;
```

---

d. Drop the Column Time from the Table

```
ALTER TABLE Reserves
DROP COLUMN Time;
```

---

e. Delete a Row with a Condition

Example: Delete the record where **Boat_id = 103**.

```
DELETE FROM Reserves
WHERE Boat_id = 103;
```

COMMIT;

Display All Records
SELECT * FROM Reserves;

SQL> SET LINESIZE 200;
SQL> SET PAGESIZE 100;
SQL> COLUMN RESERVE_DATE FORMAT A15;